



THE ASIAN SCHOOL, DEHRA DUN

Worksheet 1 — Exploring mixtures and their separations

Class : IX

Subject : Chemistry

MM : 20

Teacher : SNG

General Instructions:

1. All questions are compulsory.
2. Marks allotted to each question are indicated against it.

Section A: Multiple Choice Questions (1 mark each)

Q.No	Question	Marks
1	Which method is used to separate camphor from sand? A. Sublimation B. Filtration C. Sedimentation D. Winnowing	1
2	A mixture of Sulphur and carbon disulphide is A. Heterogeneous and show Tyndall effect B. Homogeneous and show Tyndall effect C. Heterogeneous and does not show Tyndall effect D. Homogeneous and does not show Tyndall effect	1
3	Which of the following can be classified as a mixture? A. A clear white salt solution B. A rusted iron nail C. A piece of paper cut into different shapes D. A bowl of water with floating ice cubes	1
4	The process of formation of cheese (paneer) from milk involves: A. Centrifugation B. Coagulation C. Sublimation D. Distillation	1
5	What is the term for the maximum amount of solute that can be dissolve in	1

	a fixed quantity of solvent at a given temperature? A. Concentration B. Saturation C. Solubility D. Density	
--	---	--

Section B: Short Answer Questions (2 marks each)

Q.No	Question	Marks
6	Which apparatus is used to separate two immiscible liquids, and on what property does it work?	2
7	While diluting a solution of salt in water, a student by mistake added acetone (boiling point 56°C). What technique can be employed to get back the acetone? Justify your choice.	2

Section C: Short Answer Questions (3 marks each)

Q.No	Question	Marks
8	What is crystallisation? Why is crystallisation preferred over evaporation for obtaining pure crystals of a solid? Give two reasons.	3
9	A solution contains 30g of glucose and 20 g of salt in 500ml of water. Calculate the mass per cent of (i) glucose, (ii) salt (density of water = 1g/ml).	3

Section D: Long Answer Question (5 marks)

Q.No	Question	Marks
10	A student is given the following mixtures ; • Salt and water • Sand and water • Ammonium chloride and common salt • Cream and milk • Different coloured pigments in ink Name the most suitable method for separating each mixture and state the principal involved.	5

Worksheet 2— Journey inside the atom

Class : IX

Subject : Chemistry

MM : 20

Teacher : SNG

General Instructions:

1. All questions are compulsory.
2. Marks allotted to each question are indicated against it.

Section A: Multiple Choice Questions (1 mark each)

Q.No	Question	Marks
1	Which element has two shell and both these shells are filled with electrons? A. S B. Ne C. N D. He	1
2	Rutherford's alpha particle scattering experiment led to the discovery of; A. Neutron B. Nucleus C. Proton D. Electron	1
3	Lithium has 3 electrons, 4 neutrons and 3 protons. Its mass number is: A. 3 B. 4 C. 7 D. 1	1
4	Which of the following elements has same number of protons, electrons and neutrons? A. Magnesium B. Beryllium C. Helium D. All of these	1
5	Which of the following does not have neutrons? A. H B. He C. Ne	1

	D. C	
--	------	--

Section B: Short Answer Questions (2 marks each)

Q.No	Question	Marks
6	According to Bohr's model, an electron does not radiate energy while moving in a particular orbit. Explain this statement and name these orbits.	2
7	(i) An atom has 13 protons and 14 neutrons. Calculate the valency and number of electron in an atom. (ii) The atomic nucleus of an element has mass number 39 and number of neutrons 20. What is the atomic number of an element? What is the symbol of an element?	2

Section C: Short Answer Questions (3 marks each)

Q.No	Question	Marks
8	In what way is the Rutherford's atomic model different from that of Thomson's atomic model?	3
9	Explain how Rutherford alpha-particle scattering experiment led to the discovery of the atomic nucleus. Mention any two properties of the nucleus.	3

Section D: Long Answer Question (5 marks)

Q.No	Question	Marks
10	A. Boron occurs in nature in the form of two isotopes having mass number 10 and 11. Calculate the percentage abundances of two isotopes in a sample having atomic mass 10.80. B. What is the relationship between two elements X and Y whose atomic number are 18 and 20 respectively but their mass numbers are the same 40. Are their chemical properties same or different? C. Two atoms have the same atomic number but different atomic numbers. What are they called? Give an example.	5